

FedCVESA: Taking Away Training Data in Federated Learning via Correlation Value Encoding and Segmented Aggregation

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Abstract. Federated learning (FL) reduces explicit data exposure by keeping raw data on local clients, but it does not eliminate privacy risks arising from the training process and the learned model itself. Meanwhile, centralized Taking Away Training Data (TATD) attacks have shown that malicious training could abuse the memorization capacity of deep models to store and later recover training data. However, this memorization-based threat has not been systematically studied under FL environments, where multi-client averaging can overwrite encoded training data. In this paper, we study a white-box TATD attack in which a malicious server targets selected clients and actively writes private training data into the global model during federated training. We propose FedCVESA, a federated variant of Correlation Value Encoding Attack (CVEA), by adding a Pearson-correlation regularizer to the loss function of target clients, so that private training data are gradually encoded into selected model-parameter indices, referred to as carrier positions. To reduce the overwriting of carrier positions during server aggregation, we further propose segmented aggregation over dispersed carrier positions, preserving selected carrier positions while keeping standard averaging on the remaining parameters. Experiments on MNIST, Fashion-MNIST, and grayscale 24×24 versions of CIFAR-10 show that the proposed method can steal semantically meaningful training images from the trained model while maintaining acceptable main-task utility in a controlled proof-of-concept setting. These results demonstrate that FL can become a parameter-level memorization channel for active TATD attack under the studied white-box malicious-server setting.

Keywords: Federated Learning · Privacy Leakage · White-Box Attack · Taking Away Training Data · Correlation Value Encoding Attack

1 Introduction

Federated learning (FL) allows multiple clients to jointly train a model without directly sharing raw training data, and has therefore become a standard

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paradigm for privacy-sensitive applications such as healthcare, finance, and mobile intelligence [16, 8, 27, 12]. However, the slogan “data never leaves the device” should not be mistaken for end-to-end privacy protection. Model updates still expose rich information about local training data, and a sufficiently capable adversary can exploit gradients or parameters to infer private data [32, 5, 19, 24, 1].

Most prior white-box data leakage attacks against FL focus on gradient inversion. Representative methods such as DLG, iDLG, and GradInversion recover input training data by matching dummy gradients to observed gradients [32, 31, 30, 7]. These attacks demonstrate that gradients leak substantial information, but they remain fundamentally passive: the attacker can only reconstruct whatever information is already exposed by the observed updates. Moreover, in federated learning, the leakage signal is easily weakened by multi-client averaging, which makes direct reconstruction harder to sustain across rounds.

Centralized Taking Away Training Data attacks study a different privacy risk: malicious training can force a model to remember training data. Early work showed that models can be intentionally trained to remember too much [22]. Subsequent attacks encoded training data through parameter combinations, exploited unused model capacity for training-data leakage in a black-box setting, hiding leakage behind backdoor-style triggers, or stored information in output confidences [13, 14, 28, 29, 18]. These studies reveal that model memorization can be turned into a data-stealing channel, but they are mainly designed for centralized settings.

To the best of our knowledge, existing FL privacy studies have not systematically examined whether model memorization can be actively exploited to take away target-client training data under federated aggregation. In FL, training data produced by target clients are mixed with updates from many non-target clients, and standard aggregation could overwrite the carrier positions before the server can recover the data. Thus, an FL Taking Away Training Data attack must coordinate local encoding, aggregation-time preservation, and server-side recovery.

We address this gap by studying a white-box TATD attack in FL. Instead of passively reconstructing data from whatever gradients happen to leak, we actively modify the local optimization objective on selected target clients so that private training data are gradually written into a dispersed subset of model parameters throughout federated training. Our method, called FedCVESA, builds on Correlation Value Encoding Attack (CVEA) [22] and adapts it to FL with a malicious-server segmented aggregation scheme that reduces overwriting of the encoded training data.

The main contributions of this paper are as follows.

- We formulate a white-box TATD threat for federated learning, in which selected target clients are subjected to active memorization-based data encoding during training and a malicious server performs post-aggregation extraction and recovery from the aggregated global model.

- We propose FedCVESA, which combines local correlation value encoding with segmented aggregation over dispersed carrier positions, so that training data can be recorded in model parameters while the model remains on the normal FL optimization path.
- We validate FedCVESA on MNIST, Fashion-MNIST, and a simplified grayscale 24×24 CIFAR-10 setting. The results show that the method can steal semantically meaningful images while maintaining acceptable main-task utility, highlighting an effectiveness–utility trade-off in this proof-of-concept setting.

2 Related Work

2.1 Data Leakage Attacks in Federated Learning

Early studies showed that collaborative learning can leak private information even when raw data never leaves the clients, including GAN-based active leakage, unintended feature leakage, and passive or active white-box inference [6, 17, 19].

Data-reconstruction attacks make this leakage more concrete. DLG, iDLG, GradInversion, and related gradient-inversion methods recover input training data by matching or regularizing observed gradients [32, 31, 30, 5]. Recent work extends this risk beyond standard image gradients: generative priors, modified models, and gradient magnification improve extraction in harder settings [7, 3, 25, 1], while SRATTA, Cocktail Party Attack, and TabLeak study recovery from aggregated or non-image FL signals [2, 15, 9, 23]. These methods remain mostly passive in the sense that they exploit information already exposed by updates or aggregated models.

2.2 Taking Away Training Data Attacks

Taking Away Training Data (TATD) attacks study how malicious training can make a learned model store and later reveal training data while keeping the main task usable. This threat is different from membership inference or model inversion [21, 4]: the attacker does not only infer whether a data point, attribute, or class representative exists, but aims to take away concrete training data from the trained model. The idea can be traced back to Song *et al.* [22], who first showed that machine learning models can be intentionally trained to remember too much through parameter-level and label-based encoding channels. More recently, Luo *et al.* [13] explicitly studied this memorization-based threat as a Taking Away Training Data attack and proposed Parameter Combination Encoding Attack (PCEA), which encodes private data through linear combinations of model parameters.

Existing TATD attacks can be grouped according to the attacker’s access to the trained model. In white-box settings, the attacker can inspect model parameters and use them as a storage channel. Song *et al.* [22] demonstrated parameter-level encoding, and PCEA further encodes training data through parameter combinations in a white-box setting [13]. Confidence Memory Attack

later encodes training data into output confidences, bridging white-box parameter access and black-box query-only access in a gray-box setting [18]. In black-box settings, the attacker cannot directly inspect parameters and instead relies on model outputs or maliciously designed data. Capacity Abuse Attack and its improved version exploit model capacity under black-box access by using malicious data points and label/output encodings to recover training data after model training [14]. Other studies improve the concealment of black-box data-leakage attacks by replacing obviously synthetic malicious data points with backdoor-triggered or imperceptibly modified data points [28, 29].

These TATD attacks demonstrate that model memorization can be abused as a data-storage channel. However, they are studied mainly in centralized, outsourced-training, or model-provider settings, where training is not subject to federated aggregation. They do not address the FL setting, where encoded training data from target clients can be overwritten by other clients' updates during aggregation. This leaves open whether active TATD can be made viable in federated learning, where the attacker must coordinate local encoding with server-side aggregation and recovery.

2.3 CVEA and Parameter-Level Encoding

Correlation Value Encoding Attack (CVEA) is the direct technical foundation of FedCVESA. Song *et al.* [22] showed that malicious training can turn model parameters into a storage channel by maximizing the statistical correlation between parameters at selected carrier positions and an attacker-chosen signal. Unlike gradient inversion, CVEA actively writes information into the model rather than waiting for incidental leakage.

CVEA is suitable for our FL setting because its differentiable correlation objective can be inserted into the local client loss, its carrier positions are an explicit set of model-parameter indices, and its strength can be controlled to study the stealing quality and utility trade-off. FedCVESA transfers this parameter-level encoding idea to federated learning, where segmented aggregation is needed to prevent encoded training data from being overwritten by multi-client averaging.

3 Threat Model

We consider a synchronous cross-device FL system with one server and K clients training a neural network $f(\cdot; w)$. Here, w denotes the full trainable parameter vector of the neural network, and w_t denotes the global parameter vector at communication round t . In each round, the server broadcasts w_t to selected clients. Each selected client performs local optimization on its private dataset and uploads an updated local parameter vector w_t^k . The server then aggregates these updates to obtain w_{t+1} . For the parameter-level encoding attack studied in this paper, θ denotes the flattened vector of parameter values extracted from selected model-parameter indices, referred to as carrier positions.

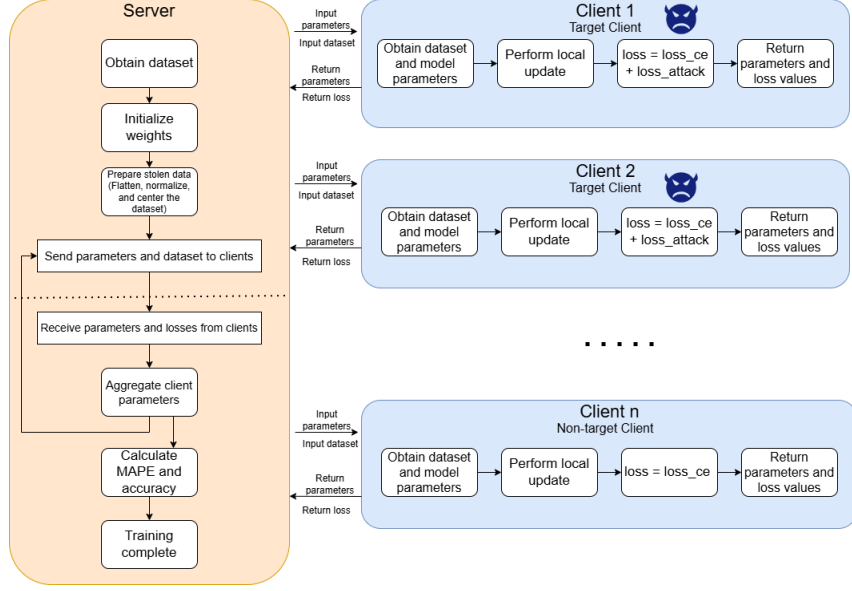


Fig. 1. Overall pipeline of the proposed white-box TATD attack in federated learning under a malicious server targeting selected clients.

The adversary is white-box and includes a malicious server that targets a small set of selected clients. We refer to these victims as target clients, and to the remaining participants as non-target clients. It knows the model architecture, training pipeline, and has access to model parameters throughout training. On target clients, the local training objective is manipulated by the attack routine, while the malicious server can customize aggregation on carrier positions before publishing the next global model. After aggregation, the malicious server can inspect the global model and run a recovery procedure. The encoded training data are drawn from private training data held locally by the target clients during normal federated learning. Under the standard FL data-isolation assumption, these raw training data are not directly accessible to the server.

The attack goal is to encode private training data during training and later recover them from the global model while preserving acceptable main-task behavior.

4 Proposed Method

FedCVESA turns federated training into a three-stage parameter-level encoding process for TATD under a malicious server targeting selected clients, as illustrated in Fig. 1. First, the attack routine performs local correlation encoding on target clients so that private training data are gradually written into carrier positions during normal local optimization. Second, the malicious server applies

segmented aggregation over dispersed carrier positions to prevent the encoded training data from being overwritten by federated averaging while leaving the remaining parameters on the standard FL update path. Third, the malicious server extracts the preserved values from the global model and rearranges them to recover the training data. The following subsections introduce local correlation encoding, segmented aggregation over dispersed carrier positions, and server-side extraction and recovery.

4.1 Local Correlation Encoding

Standard client training minimizes a classification loss L_{ce} . To turn local training into a parameter-encoding process, we augment the local objective on target clients with a correlation regularizer:

$$L_{total} = L_{ce} - \gamma L_{corr}, \quad (1)$$

where γ controls attack strength. Following the CVEA principle introduced in Section 2.3, L_{corr} is the federated instantiation of the malicious correlation term and encourages the model parameters to become statistically aligned with private training data.

In our implementation, the encoded training data are derived from private grayscale training images on the target clients after unified preprocessing. Consistent with this notation, let θ denote the flattened carrier-position vector used for attack carrying, and let d denote the flattened training-data vector. Both vectors have the same length, and the subscript i indexes their corresponding elements. The terms $\bar{\theta}$ and $\bar{d}$ denote the empirical means of θ and d , respectively. We instantiate the CVEA-style malicious term with Pearson-style correlation:

$$L_{corr} = \frac{\sum_i (\theta_i - \bar{\theta})(d_i - \bar{d})}{\sqrt{\sum_i (\theta_i - \bar{\theta})^2} \sqrt{\sum_i (d_i - \bar{d})^2}}. \quad (2)$$

Maximizing this term gradually biases the optimization trajectory so that training data are written into the parameter space. Smaller γ better preserves main-task utility but yields weaker encoding; larger γ improves the fidelity of the stolen images but risks disrupting the main task.

4.2 Segmented Aggregation over Dispersed Carrier Positions

The attack requires a sufficient number of carrier positions to record training data. A naive design is to use a contiguous parameter block. However, when too many carrier positions are contiguous, they may disproportionately perturb one or more network layers, and then could distort that layer’s normal representation and noticeably degrade the main task. Therefore, we do not use a contiguous block of carrier positions. Instead, we flatten the full model and construct a set of carrier-position indices across the entire parameter vector:

$$s_r = \left\lfloor \frac{r(L-1)}{T-1} \right\rfloor, \quad r = 0, 1, \dots, T-1, \quad (3)$$

where L is the flattened parameter length and T is the total number of carrier positions. These indices are then logically partitioned into client-specific subsets. This design does not increase the encoding budget; it disperses a fixed number of carrier positions across the model, preventing the encoding burden from being concentrated in any single layer.

During server aggregation, let $\mathcal{C}_t$ denote the clients selected in round t , and let $\mathcal{C}_{\text{target}} \subseteq \mathcal{C}_t$ denote the target clients among them. For each target client $c \in \mathcal{C}_{\text{target}}$, Ω_c denotes the carrier-position index set assigned to that client. These client-specific sets are disjoint, and their union is the full carrier-position set:

$$\Omega_{\text{attack}} = \bigcup_{c \in \mathcal{C}_{\text{target}}} \Omega_c, \quad \Omega_c \cap \Omega_{c'} = \emptyset \ (c \neq c'). \quad (4)$$

The non-carrier positions are the complement of Ω_{attack} in the full flattened parameter-index set and still follow standard averaging. The malicious server preserves target-client values on the corresponding carrier positions and applies standard averaging elsewhere:

$$w_{t+1}(j) = \begin{cases} w_t^c(j), & j \in \Omega_c, \ c \in \mathcal{C}_{\text{target}}, \\ \frac{1}{|\mathcal{C}_t|} \sum_{k \in \mathcal{C}_t} w_t^k(j), & j \notin \Omega_{\text{attack}}. \end{cases} \quad (5)$$

Because the sets $\{\Omega_c\}_{c \in \mathcal{C}_{\text{target}}}$ are disjoint, the first case has a unique target client c for each carrier position $j \in \Omega_{\text{attack}}$. This segmented aggregation preserves selected carrier positions while keeping the model on the normal FL optimization path. Together with dispersed carrier-position placement, it mitigates both aggregation-induced overwriting of the encoded training data and local disruption caused by concentrating too many carrier positions in a single layer.

4.3 Server-Side Extraction and Recovery

After aggregation, the malicious server extracts the preserved carrier positions from the global model and decodes them using the same index order as in local encoding and segmented aggregation. Using the previously defined client-specific carrier-position set Ω_c , the server extracts the encoded vector:

$$z_c = [w_{t+1}(j)]_{j \in \Omega_c}, \quad c \in \mathcal{C}_{\text{target}}. \quad (6)$$

For a single training image with preprocessed size $H \times W$, we set $|\Omega_c| = S = HW$ and recover the image by min-max normalization and reshaping:

$$\begin{aligned} \hat{x}_c &= \text{Reshape}(\text{Norm}(z_c), H, W), \\ \text{Norm}(z_c)_i &= \frac{z_{c,i} - \min(z_c)}{\max(z_c) - \min(z_c) + \epsilon}. \end{aligned} \quad (7)$$

Here, $\text{Norm}(\cdot)$ maps extracted parameter values to the grayscale range, and ϵ avoids numerical instability when the extracted values have a very small range.

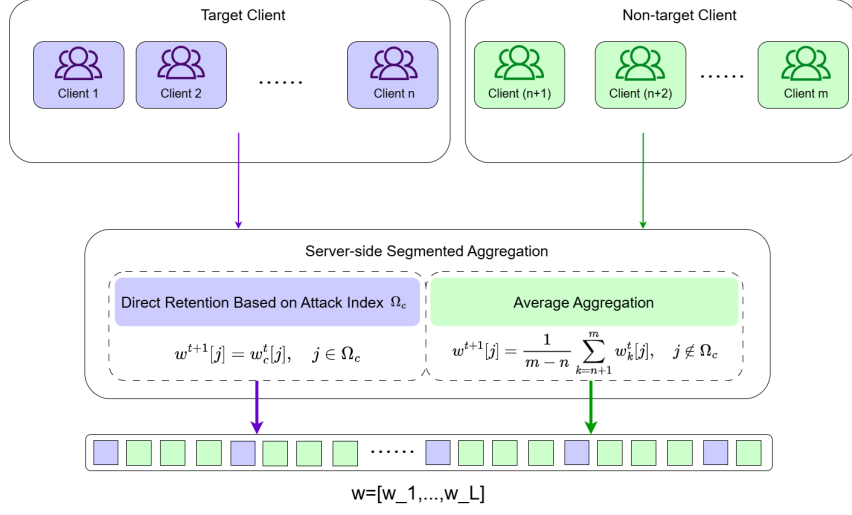


Fig. 2. Segmented aggregation with dispersed carrier positions. Logical segments are assigned to target clients, but the carrier positions are globally scattered in the parameter vector.

This recovery step does not require an additional optimization process; it depends on using the same index rule, encoding length, and reshaping convention across local encoding, segmented aggregation, and server-side extraction. The resulting objective is not perfect pixel-wise recovery, but semantically meaningful stolen images under the studied FL aggregation setting.

5 Experiments

5.1 Experimental Setup

We implement FedCVESA in a PyTorch-based federated learning simulator [20]. The experiments are conducted on MNIST, Fashion-MNIST, and CIFAR-10 [11, 26, 10]. All datasets are partitioned in an IID manner across clients. MNIST and Fashion-MNIST are used as normalized grayscale datasets without cropping, while CIFAR-10 images are converted to grayscale, cropped to 24×24 , and normalized before training. Therefore, the CIFAR-10 results should be interpreted as a simplified grayscale 24×24 setting rather than the original color benchmark.

Unless otherwise specified, we use 100 total clients, client fraction 0.1, 200 communication rounds, and segmented aggregation. The attack warm-up length is set to 100 rounds: for $\gamma > 0$, the first 50 rounds use $\gamma = 0$, followed by a 50-round cosine ramp to the target value. The default setting uses 5 target clients and one private training image per target client. We use a CNN for MNIST and Fashion-MNIST. For the simplified CIFAR-10 setting, we report two model configurations: CIFAR-CNN denotes CIFAR-10 trained with the CNN,

and CIFAR-ResNet18 denotes CIFAR-10 trained with ResNet18. Local epochs are set to 10 for MNIST and Fashion-MNIST, and 5 for CIFAR-10; batch size is set to 10 for MNIST and Fashion-MNIST, and 50 for CIFAR-10.

We evaluate FedCVESA from both utility and stealing perspectives. Utility is measured by final test accuracy and final loss at round 200. Stealing quality is measured by mean absolute percentage error (MAPE) between reconstructed training data and the corresponding private training data; lower MAPE indicates better stealing quality. For training data x and its reconstruction $\hat{x}$ with n pixels, we compute

$$\text{MAPE}(x, \hat{x}) = \frac{1}{n} \sum_{i=1}^n \frac{|x_i - \hat{x}_i|}{|x_i| + \epsilon}, \quad (8)$$

where ϵ is a small constant used to avoid division by zero for near-black pixels. For the no-attack baseline ($\gamma = 0$), MAPE is not reported because no training data are encoded.

The reported experiments focus on three questions. First, we vary the attack strength over $\gamma \in \{0, 0.05, 0.2, 0.5, 1.0\}$ while fixing five target clients, one training image per client, and segmented aggregation; this study covers MNIST, Fashion-MNIST, CIFAR-CNN, and CIFAR-ResNet18. Second, we vary the number of target clients over $\{1, 2, 3, 4, 5, 10\}$ while fixing $\gamma = 0.5$, one training image per client, and segmented aggregation; this study covers MNIST, Fashion-MNIST, and CIFAR-CNN. Third, we ablate the carrier-position placement strategy under a higher-payload setting, comparing contiguous placement with the proposed dispersed placement while fixing $\gamma = 0.5$, five target clients, ten training images per client, and segmented aggregation.

5.2 Attack Strength Analysis

The first study varies the attack strength γ while fixing five target clients, one training image per target client, and segmented aggregation. Table 1 reports the final accuracy, loss, and MAPE at round 200, while Fig. 3 shows the final utility–stealing trade-off.

On MNIST and Fashion-MNIST, main-task utility remains stable across attack strengths, while the best MAPE appears at moderate or low γ values rather than monotonically improving with stronger attacks. This indicates that increasing γ alone is not a reliable way to improve stealing quality on simpler or texture-rich grayscale training data.

The simplified CIFAR-10 setting is more sensitive. For CIFAR-CNN, attacked models lose about 3–4.5 accuracy points relative to the 72.86% baseline, while MAPE decreases from 0.1766 at $\gamma = 0.05$ to 0.0752 at $\gamma = 1.0$, showing a clearer utility–stealing trade-off.

Comparing CIFAR-CNN with CIFAR-ResNet18 further shows that model structure matters: in the current configuration, the larger architecture has lower accuracy and its best MAPE appears at $\gamma = 0.5$, so model size alone does not guarantee a better carrier-position set.

Table 1. Attack strength scan under five target clients and one training image per client.

Dataset/model	γ	Acc	Loss	MAPE
MNIST	0	99.05%	0.3384	–
	0.05	99.03%	0.3428	0.2265
	0.2	99.02%	0.3394	0.2109
	0.5	99.17%	0.2912	0.2967
	1.0	99.05%	0.3251	0.2606
Fashion-MNIST	0	92.67%	3.1411	–
	0.05	92.23%	3.1534	0.1298
	0.2	92.28%	3.1171	0.1878
	0.5	92.20%	3.3485	0.2552
	1.0	92.48%	3.3907	0.1393
CIFAR-CNN	0	72.86%	8.6448	–
	0.05	68.36%	11.1172	0.1766
	0.2	69.54%	10.7524	0.1047
	0.5	69.70%	10.7609	0.0789
	1.0	69.86%	10.6049	0.0752
CIFAR-ResNet18	0	68.76%	10.5448	–
	0.05	65.94%	12.7644	0.1795
	0.2	65.88%	12.6463	0.1279
	0.5	66.66%	12.5741	0.1030
	1.0	66.06%	12.2854	0.1384

5.3 Target-Client Scale Analysis

The second study fixes $\gamma = 0.5$, one training image per target client, and segmented aggregation, and varies the number of target clients. This setting evaluates whether FedCVESA remains stable as the amount of simultaneously stolen private training data increases.

To make the scale effect easier to compare across datasets, Fig. 4 plots final accuracy and MAPE against the number of target clients.

For MNIST and Fashion-MNIST, increasing the number of target clients from 1 to 5 has little effect on classification accuracy; at 10 target clients, both datasets show larger utility degradation. The best MAPE values do not always coincide with the largest acceptable utility: 3 or 4 target clients is more balanced on MNIST, and 3 is most favorable on Fashion-MNIST.

CIFAR-CNN is more sensitive: moving from 1 to 10 target clients improves MAPE from 0.2157 to 0.0734 but reduces accuracy from 72.84% to 59.96%. Thus, 2, 3, or 5 target clients provides a more practical compromise.

5.4 Carrier-Position Placement Ablation

The third study examines whether the dispersed carrier-position design is useful when the attack payload becomes larger. We fix $\gamma = 0.5$, five target clients, ten

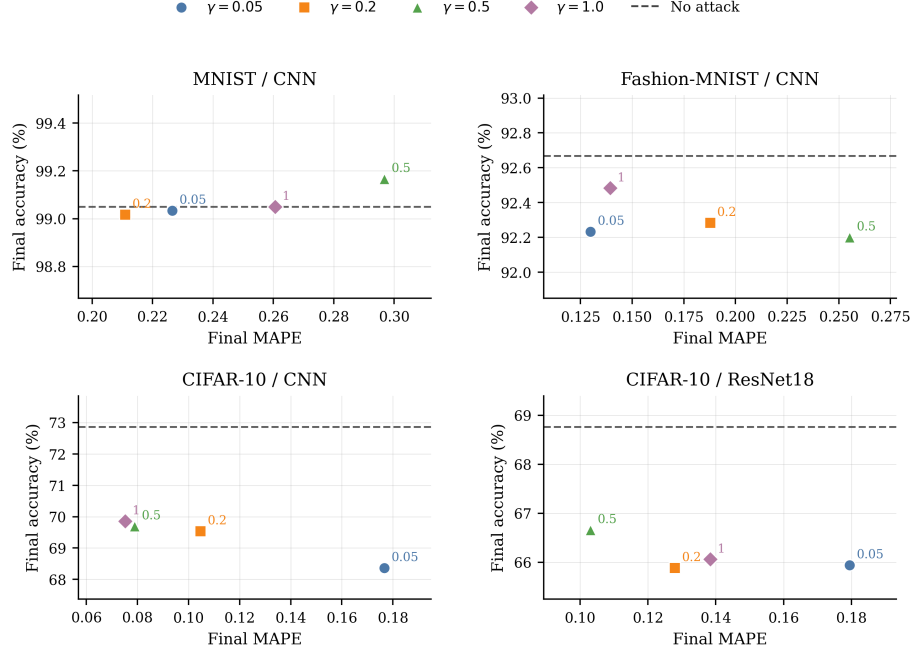


Fig. 3. Final utility-stealing trade-off under different attack strengths. Points denote γ values; the dashed line is the no-attack accuracy baseline.

training images per target client, and segmented aggregation. The total number of carrier positions follows the dataset-specific preprocessed image size: $5 \times 10 \times 784 = 39200$ positions for MNIST and Fashion-MNIST, and $5 \times 10 \times 576 = 28800$ positions for the simplified 24×24 CIFAR-10 setting. We compare two placement strategies: *contiguous placement*, which uses the first N carrier positions of the flattened model parameter vector, and *dispersed placement*, which uniformly

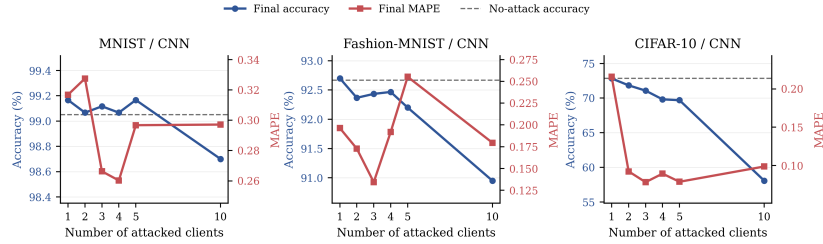


Fig. 4. Effect of target-client scale on final accuracy and MAPE. The dashed line marks the no-attack accuracy baseline.

Table 2. Target-client scale scan under $\gamma = 0.5$ and one training image per client.

Dataset/model	Target clients	Acc	Loss	MAPE
MNIST	1	99.17%	0.3328	0.3168
	2	99.07%	0.3289	0.3275
	3	99.12%	0.3117	0.2663
	4	99.07%	0.3154	0.2603
	5	99.17%	0.2912	0.2967
	10	98.53%	0.6561	0.2375
Fashion-MNIST	1	92.70%	3.2258	0.1964
	2	92.37%	3.2539	0.1727
	3	92.43%	3.2606	0.1342
	4	92.47%	3.2421	0.1919
	5	92.20%	3.3485	0.2552
	10	90.90%	5.0797	0.2404
CIFAR-CNN	1	72.84%	8.7642	0.2157
	2	71.84%	9.5343	0.0924
	3	71.08%	9.4305	0.0784
	4	69.82%	10.3484	0.0895
	5	69.70%	10.7609	0.0789
	10	59.96%	21.0074	0.0734

samples N dispersed carrier positions across the full flattened parameter vector according to Eq. 3.

Table 3 shows that dispersed placement is consistently preferable in this high-payload setting. It obtains higher accuracy and lower loss across all datasets, while also improving MAPE on MNIST and Fashion-MNIST and slightly improving it on CIFAR-CNN. These results suggest that distributing carrier positions across the model reduces the damage caused by concentrating many carrier positions in a contiguous functional region.

5.5 Discussion

Overall, FedCVESA is governed by an effectiveness–utility trade-off shaped by attack strength, target-client scale, carrier-position placement, dataset complexity, and model structure. MNIST and Fashion-MNIST tolerate moderate attacks with little accuracy loss, whereas CIFAR-CNN obtains clearer MAPE improvements from stronger or larger-scale attacks at a higher utility cost. The CIFAR-ResNet18 and placement-ablation results further show that larger architectures and concentrated carrier positions do not automatically improve stealing quality; dispersed placement is more stable under larger payloads.

Increasing γ strengthens each piece of encoded training data, whereas increasing the number of target clients expands the encoded training data and protected carrier-position subsets. Both can reduce MAPE in some settings, but the gain is meaningful only while main-task accuracy remains acceptable.

Table 3. Carrier-position placement ablation under $\gamma = 0.5$, five target clients, and ten training images per client.

Dataset/model	Placement	Acc	Loss	MAPE
MNIST	Contiguous	98.75%	0.3939	0.1617
	Dispersed	99.05%	0.3326	0.1390
Fashion-MNIST	Contiguous	91.60%	3.3675	0.1938
	Dispersed	92.32%	3.2829	0.0655
CIFAR-CNN	Contiguous	66.92%	11.4358	0.1381
	Dispersed	69.04%	10.6310	0.1363

6 Conclusion

This paper studies a white-box TATD attack in federated learning. By adapting CVEA to the FL setting, FedCVESA turns local optimization into an active memorization process and uses segmented aggregation over dispersed carrier positions to reduce overwriting of encoded training data. Experiments on MNIST, Fashion-MNIST, and the simplified grayscale 24×24 CIFAR-10 setting show that the proposed method can steal semantically meaningful private training data while preserving acceptable task utility in a range of settings. These findings provide proof-of-concept evidence indicating that, under strong white-box adversaries, FL may function as a parameter-level memorization channel, though this should not be interpreted as a comprehensive assessment of real-world risk.

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